Shriya Garlapati

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PROFESSIONAL SUMMARY

Software Engineer with 3+ years of experience in software engineering and applied ML research, specializing in scalable full-stack systems (Java, Spring Boot, React) and event-driven architectures. Skilled in cloud infrastructure, microservices, and applied ML, including XGBoost and RAG-based GenAI solutions.

WORK EXPERIENCE

Software Engineer Jun 2026 - Present

Service Oriented Solutions LLC, Remote, USA

- Architected a full-stack release management platform using **Spring Boot and React**, replacing an error-prone process and cutting release processing time by **40%** while reducing errors by **60%**.
- Integrated third-party **transportation management APIs** to automate order and shipment workflows, improving transaction accuracy by **50%** and eliminating manual data entry by **70%**.

Research Assistant – Software Engineering Aug 2025 - May 2026

University of Georgia, Remote, USA

- Built a **RAG-based GenAI microservice** using **LangChain** and **GPT APIs** to retrieve user-specific ML history and generate structured explanations of crop yield predictions.
- Engineered a **fault-tolerant Kafka** consumer persisting ML prediction results to **PostgreSQL** with **post-persistence manual offset commits**, **idempotent deduplication**, and **dead-letter topic routing**, minimizing data loss under transient failures.
- Decoupled email delivery from the signup critical path using **async Kafka** pipeline, **reducing p95 latency by ~350ms.**
- Designed a stateless **JWT-based authentication** and **RBAC authorization** for a **distributed microservices** platform, centralizing token issuance and validation while standardizing authorization across services.
- Built a centralized JWT auth layer with **React Context API** and **Axios interceptors**, consolidating token attachment, refresh, and session handling.

Graduate Researcher Aug 2024 - May 2025

University of Georgia, Athens, GA

- Developed an **end-to-end ML pipeline** forecasting alfalfa biomass yields across 4 NY locations using 20 years of agricultural trial data (7,344 records), achieving a best **R^2 of 0.91 and SMAPE of 2.99%** via **Optuna-tuned XGBoost**.
- Improved model performance through **Optuna Bayesian hyperparameter search** and ensemble methods (**voting and stacking regressors**), **boosting R^2 by up to 10.5%** and **cutting SMAPE by up to 11.6%** across four cities.
- Devised a two-stage augmentation pipeline using uniform perturbation and **SMOTE** to transform aggregated agricultural samples into temporally balanced training records for regression modeling.
- Presented *Ensemble Learning for Alfalfa Biomass Yield Forecasting* as an accepted poster at the **2026 AI in Agriculture Conference**, NC State University, Raleigh, NC.

Software Engineer I (Full Stack Developer) Jan 2022 - Jul 2023

NCR Voyix, Hyderabad, India

- Reduced NCR Counterpoint API latency by **93%** by engineering a **Redis caching layer** with **multi-tenant namespace isolation**, reducing redundant database reads for pay codes, tax codes, and gift card codes on every transaction.
- Integrated **Elasticsearch** as a search read model, enabling **sub-20ms full-text search** with **fuzzy matching** and relevance ranking improving the latency by **10x** over full-table-scans.
- Optimized the customers endpoint using **JPA pagination** and **DTO projections** on lazy-loaded collections, preventing **N+1 queries** and reducing risk of connection pool exhaustion under heavy sync load.
- Built the User Access Control and Role Management modules for the admin console using **TanStack Query** and **Zustand**, with transactional role assignment, zero-role user warnings, and domain-grouped permission picker.
- Authored comprehensive **JUnit and Mockito** unit tests, establishing an **80% code coverage** guardrail for the deployment pipeline.

Software Engineer Intern May 2021 - Jul 2021

S&P Global, Hyderabad, India

- Architected 15+ **Cloud Custodian** policies using YAML, **CloudTrail**, and SQS across **8 AWS services** (EC2, S3, IAM, DynamoDB, ECS, ECR, Lambda, DMS) to automate compliance enforcement and reduce **security drift by 90%**.
- Implemented tagging, IAM, and ECR governance policies in **Cloud Custodian**, reducing **non-compliant resources by 95%** and improving cost attribution.

TECHNICAL SKILLS

Programming: Java, Python, JavaScript, TypeScript, SQL, HTML, CSS
Frameworks & Technologies: Spring Boot, Spring Security, React, Redux, Apache Kafka, Spring Data JPA, Hibernate, REST APIs, OAuth 2.0
Databases: PostgreSQL, MongoDB, MySQL, Oracle, Redis, Elasticsearch
Cloud & DevOps: AWS, Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, CI/CD Pipelines, Prometheus, Grafana
Testing & Build Tools: JUnit 5, Mockito, Maven, Postman, OpenAPI 3.0 (Swagger)
AI/ML: Scikit-learn, XGBoost, PyTorch, Optuna, LangChain, Time Series Forecasting, Data Augmentation & Resampling
Developer Tools & Methodologies: Microservices, Event-driven Architecture, System Design, Git, Jira, Agile, Scrum
Certifications: Microsoft Certified: Azure Fundamentals | AWS Cloud Practitioner | IBM Full Stack Software Developer Professional Certificate
Memberships & Conferences: Grace Hopper Celebration (GHC 2026) Registered Attendee, AnitaB.org member

EDUCATION

Master of Science in Computer Science, University of Georgia, CGPA: 3.85 / 4.0 — Aug 2023 - May 2025
Relevant Coursework: Data Science, Algorithms, Database Management, Distributed Computing Systems
Bachelor of Technology, Computer Science, G. Narayanamma Institute of Technology & Science — Aug 2018 - Jul 2022